

PhD Research Proposal

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Dynamic Transfer Learning for Low-Resource Sequence-to-Sequence Models

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Abstract

Our research proposes a dynamic transfer learning approach for low-resource NMT that uses parent knowledge during child inference accordingly. Although, kNN-TL came up to integrate parent knowledge during all three phases (child initialization, training, and inference), it lacks the dynamic aspect as well as suffer from inefficiencies given the expensive computation and storage costs. We aim to develop a confidence-aware framework that selectively resorts to parent knowledge only when the child’s predictions are uncertain. Such an approach intends to improve translation quality while reducing latency. The proposed method will later be evaluated of diverse low-resource language pairs, giving out a practical solution to ameliorate inference-time efficiency and effectiveness in various seq2seq tasks, opening doors to possible generalization beyond machine translation.

Contents

1	Motivation, Goal & Problem Statement	4
1.1	What are your main research objectives?	4
1.2	Why is this research important?	4
1.3	What difference do you think the research will make?	4
1.4	What research gaps will you be filling?	4
1.5	How might your research add value to the subject?	5
1.6	Is your research achievable in the time allocated (e.g., 3 years)?	5
2	Literature Review	5
2.1	Low-resource NLP	5
2.2	Neural Machine Translation	6
2.3	Transfer Learning for Neural Machine Translation:	6
2.4	Dynamic Transfer learning in NMT	7
2.4.1	k -Nearest-Neighbor Retrieval	7
2.5	Bridging Transfer learning and k NN-MT	8
3	Theoretical framework and Methodology	9
3.1	Theoretical Framework	9
3.2	Methodology and techniques	10
4	Preliminary Results	11
4.1	Small-scaled experiments:	11
4.2	Large-scaled experiments:	12
5	References	13

List of Figures

List of Tables

1 Motivation, Goal & Problem Statement

1.1 What are your main research objectives?

The main goal of this research is to enhance the efficiency and performance of transfer learning in our chosen low-resource seq2seq task that is **machine translation (MT) from Moroccan dialect (DRJ) to Modern Standard Arabic (MSA)** by developing a ***dynamic inference framework*** that adaptively integrates the parent model based on the child model confidence levels. Thus, We have the following research objectives fixed:

- Uncover the possible inefficiencies of static parent model usage during child inference, especially in kNN-TL.
- Develop a dynamic approach that calls for the parent model only when the child model’s confidence is low
- Evaluate the developed approach in terms of computational cost and translation quality.

The objectives mentioned above result in the following questions:

- Can a confidence-based mechanism improve the efficiency of inference-time parent model usage in low-resource MT?
- To what extent can dynamic transfer learning improve BLEU scores and reduce computational cost compared to kNN-TL?
- What is the optimal threshold of a child model’s confidence for invoking the parent model?

1.2 Why is this research important?

Balancing the effectiveness of parent model knowledge with inference-time efficiency presents itself as a challenge to tackle in machine translation. Our research aims to enable more sustainable and scalable machine translation systems, especially in low-resource settings, and perhaps develop a generalizable inference-time strategy to be applied to other seq2seq tasks.

1.3 What difference do you think the research will make?

A dynamic inference method in transfer learning for low-resource NMT preaches for a:

- Reduction in latency and computational costs during child inference.
- Improvement in translation quality.
- Setting a precedent for ‘confidence-aware inference methods’ in transfer learning for low-resource machine translation.

1.4 What research gaps will you be filling?

After thorough exploration and careful consideration, we mean to fill the following gaps:

1. Scarcity of approaches that *selectively* call for the parent model during inference based on child model confidence.
2. Absence of adaptive methods that tackle the trade-off between accuracy and speed in transfer learning for low-resource MT.

1.5 How might your research add value to the subject?

The research aims for a practical, dynamic and adaptive framework for transfer learning in low-resource NMT that meets the primary requirements of low latency along with efficient accuracy.

1.6 Is your research achievable in the time allocated (e.g., 3 years)?

Yes, I am confident to say that the research work is achievable within the time allocated, with the following milestones marking each year:

1. In-depth literature review, baseline reproduction of k NN-TL and thorough analysis of the results.
2. Developing a dynamic inference method, experimentation on our low-resource MT tasks (Moroccan dialect to MSA, All Maghrebi dialects to MSA) and ablation studies.
3. Evaluation on diverse datasets, optimization of confidence thresholds and publication of results.

As the project uses existing toolkits such as (e.g., Fairseq (a sequence modeling toolkit that allows researchers and developers to train custom models for multiple generation tasks including MT.) and public datasets (e.g., HPLT-v1, PADIC), the research tends to be feasible as long as a high-resource workstation is provided.

2 Literature Review

This is a concise comprehensive literature review of low-resource NLP, transfer learning and k -NN retrieval for low resource NMT.

2.1 Low-resource NLP

When talking about low-resource NLP¹, it's primarily related to the data and it's primordial to get the dimension of the resource's availability to cherry-pick the right method to handle the low-resource issue. Basically, it englobes:

- **Task-specific labeled data:** In need of domain expertise for a time-consuming and cost-exhaustive manual annotation of instances.
- **Unlabeled Language- or domain-specific text:** Can be trickier to obtain in some resource-lean scenarios. For example, given social media being a source for data collection, there is the issue where a large population with a mother tongue X prefer to use another language Y when using social media leaving the former language at disadvantage.
- **Auxiliary data:** Seen a support for some low-resource approaches such as labeled data in another language, a knowledge base or a machine translation tool.

¹<https://towardsdatascience.com/a-visual-guide-to-low-resource-nlp-d7b4c7b1a4bc>

2.2 Neural Machine Translation

- A standard NMT model, typically a fully parametric sequence-to-sequence model, has the goal of learning **conditional probability** $p_{NMT}(Y|X)$ to translate a sentence $X = \{x_1, \dots, x_m\}$ in source language to a sentence $Y = \{y_1, \dots, y_m\}$ in target language.
- Translation is a process done in an **autoregressive manner**, leading to its probability to be factored as :

$$p_{NMT}(Y|X) = \prod_{i=1}^n p(y_i|X, Y_{<i})$$

- When predicting the i^{th} token in the target sentence given $(X, Y_{<i})$ the **translation context**, The NMT model encodes the translation context into a hidden state h_{i-1} and outputs a probability distribution over vocabulary $\mathcal{V}$ as:

$$p_{NMT}(y_i|X, Y_{<i}) = \frac{\exp(o_{y_i}^\top h_{i-1})}{\sum_{w \in \mathcal{V}} \exp(o_w^\top h_{i-1})}$$

where o_y is the output embedding for $w \in \mathcal{V}$.

- Given $Y^* = y_1^*, \dots, y_n^*$ our ground-truth target sentence, and for each y_i^* token in the training set, its usually the **Cross-Entropy (CE) loss** that is used for the NMT models' optimization, as follows:

$$\mathcal{L}_{CE} = -\sum_{y_i \in \mathcal{V}} 1_{y_i=y_i^*} \log p_{NMT}(y_i|X, Y_{<i}^*)$$

where 1 is the indicator function, and the ground-truth target sequence $Y_{<i}^*$ is used in the conditions of p_{NMT} due to the **teacher forcing technique**, which is the act of providing the model the correct output of training data as *input* at each step rather than letting it generate the next token based on its previous outputs.

2.3 Transfer Learning for Neural Machine Translation:

Parameter inheritance is the main foundation of transfer learning in NMT, whereas a low-resource language model (child) have their parameters initialized from a high-resource language model (parent). [Zoph et al., 2016] were the first to come up with such an approach proving that parent models "teach" both target-language linguistic patterns and general translation capabilities to child models. Later on, [Kocmi and Bojar, 2018] came to reinforce the findings of [Zoph et al., 2016] and explored the simple concept of "trivial" transfer by demonstrating that even with unrelated language pairs, fine-tuning remains a strong starting point. In general, the process is explained below:

- **Parameter Initialization:** Given Θ^P the pre-trained parameters of the parent model, Θ^c the parameters of the child model, the latter is initialised by the former :

$$\Theta^c = \mathcal{I}(\Theta^P)$$

with $\mathcal{I}$ being the initialization method that either includes a part or the whole pre-trained set of parameters.

- **Fine-tuning:** The child model is to be trained on the low-resource child data (x^c, y^c) , starting from the pre-initialized parameters. The child is optimized by minimizing the cross-entropy (CE) loss function:

$$\mathcal{L}_{CE} = -\sum_{t=1}^T \log(p(y_t^c|x^c, y_{<t}^c, \Theta^c))$$

where T is the length of the target sentence, x^c is the input token, $y_{<t}^c$ is the sequence of previous outputs (before time step t).

2.4 Dynamic Transfer learning in NMT

- **Issue?**

As traditional transfer learning techniques in NMT involve parameter initialization from a pretrained high-resource "parent" model followed by fine-tuning on a low-resource "child" dataset, their common static aspect labels them as incapable of fully benefiting from the parent knowledge during the child training.

- **Key contribution!**

Consist-TL [Li et al., 2022] came up with a continuous transfer learning approach. First, for each child training instance, it creates a semantically-equivalent instance for the parent model resulting from back-translating the child target data. Second, while training the child, already initialized using the parent's parameters, Consist-TL encourages prediction consistency between both models, during each child training instance.

Tested across five low-resource language pairs, the method showcased considerable improvement in BLEU score [Papineni et al., 2002], all from engaging richer parent flow from parent to child during training.

2.4.1 k -Nearest-Neighbor Retrieval

Since 2020, following the success of non-parametric retrieval-augmented methods in NLP, for NMT, **k NN-MT** [Khandelwal et al., 2021] arised and managed to incorporate external knowledge into machine translation systems using k NN retrieval from a datastore built using the training data. In fact, it calls for the creation of a key-value datastore from encoder hidden states (translation contexts) and target tokens, the retrieval of similar translation contexts during inference, and interpolating the retrieval distribution with the model softmax. It did improve domain adaptation at inference time, but with **high latency**. The process is described below:

- **Datastore Construction:** Stores key-value pairs $(k, v) = (f(X, y_{1:i-1}), y_i)$. with $f(\cdot)$ is the representation of the translation context (X is the source sentence, $y_{1:i-1}$ are the tokens generated so far) and y_i is the target next token.
- **Retrieval-Augmented Decoding:** Given the current input source sentence X^* , at step i , the model:
 1. Computes the high-dimensional representation of current translation context, defined as query $q = f(X^*, y_{1:i-1})$.
 2. Retrieves the k nearest translation contexts from $\mathcal{S}$ along with their corresponding target tokens $\mathcal{N} = \{k_j, v_j\}_{j=1}^k$ using q and L_2 distance.
 3. Interpolates the NMT output with a k NN- based distribution:

$$p(y_i|X^*, y_{1:i-1}) = \lambda p_{kNN}(y_i|X^*, y_{1:i-1}) + (1 - \lambda)p_{NMT}(y_i|X^*, y_{1:i-1})$$

where p_{kNN} is derived from the retrieved examples (softmax over distances).

As high latency, inevitably, tagged along high translation quality, there have been works that built on the bases founded by [Khandelwal et al., 2021] tackling the issue of demanding computational cost, such as:

- **Efficient Cluster-Based k NN-MT:** [Wang et al., 2022] it managed to reduce both memory and latency costs using compressed representations and cluster-based pruning of datastore entries with minimal BLEU [Papineni et al., 2002] loss.
- **Subset-retrieval k NN-MT** [Deguchi et al., 2023]: it first selects semantically relevant sentence subsets and the nearest neighbors retrieval is done only withing them, which improved k NN-MT decoding speed.
- **Nearest Neighbor Knowledge Distillation** [Yang et al., 2022]: it calls for the transfer of the retrieval behavior of k NN-MT statically during training only, and leaving the decoding to be run as a typical NMT model would do.

2.5 Bridging Transfer learning and k NN-MT

- **k NN-TL** [Liu et al., 2023] extends the dynamic transfer learning mechanism proposed by [Li et al., 2022] to leveraging parent knowledge during child inference.
- During child training, the approach employs a parent-child representation alignment method that ensures consistency in the output latent representations of the two models.

- **Pseudo-Parent Data Construction:**

We first need to generate **pseudo parent source sentences** for the entire child data X^c, Y^c . For each instance (x^c, y^c) , a well-trained reversed parent model is employed to *back-translate* the target sentence y^c to a pseudo parent source sentence $\tilde{x}_p$ and have a pseudo parent data $(\tilde{x}^p, y^c) \in (\tilde{X}^p, Y^c)$.

- **Parent-Child Representation Alignment:**

Next comes the child training. For each instance of the pseudo parent data $(\tilde{x}^p, y^c) \in (\tilde{X}^p, Y^c)$, the parent model gives out the output representation as $f_{\Theta^p}(\tilde{x}^p, y_{<t}^c)$ for every target token y_t^c , whereas the child model generates the output representation as $f_{\Theta^c}(x^c, y_{<t}^c)$ for the same target token. We minimize the squared Euclidean distance of these two output representations with the MSE loss:

$$L_{MSE} = \sum_{t=1}^T \|f_{\Theta^p}(\tilde{x}^p, y_{<t}^c) - f_{\Theta^c}(x^c, y_{<t}^c)\|^2$$

where Θ^p, Θ^c the parent model parameters, the child model parameters respectively and T is the target sentence’s length. The final loss is a combination of the CE loss and the MSE loss, with a balancing hyper-parameter α as:

$$\mathcal{L}(\Theta^c) = \mathcal{L}_{CE}(\Theta^c) + \alpha \mathcal{L}_{MSE}(\Theta^c)$$

- During child inference, a child-aware datastore, built by selectively distilling the parent datastore based on relevance to the child model, is consulted for the k NN retrieval followed by the interpolation of both the child’s output and the retrieved distribution.

- **Child-Aware datastore construction:**

We start by **constructing the parent datastore**. Given the parent model’s data (X^p, Y^p) , we have:

$$(K, V) = \bigcup_{(x^p, y^p) \in (X^p, Y^p)} \{(f_{\Theta^p}(x^p, y_{<t}^p)y_t^p), \forall y_t^p \in y^p\}$$

Where K is the key (parent model output representations) and V is the value (golden target tokens). This generate a big datastore that stores all useful translations done by the parent model.

- We then move on to **retrieving the relevant entries** with k NN from the parent datastore using the parent output representation $f_{\Theta_P}(\tilde{x}^p; y_{<t}^c)$ for each target token y^c , given the pseudo input $\tilde{x}^p$:

$$N_{y^c} = \{(k_j, v_j), j \in \{1, 2, \dots, \tilde{k}\}, \forall y_t^c \in y^c\}$$

At the end, we merge all those relevant entries in one **Child-Aware datastore** : $(K, V) = \{N_{y^c}, \forall(\tilde{x}^p, y^c) \in (\tilde{X}^p, Y^c)\}$

- **Parent-Enhanced Model Prediction:**

During Inference, the child model outputs an intermediate representation $f_{\Theta^c}(x^c; y_{<t}^c)$ used to query the child aware datastore and get the following retrieval distribution:

$$p_{parent-kNN}(y_t^c | x^c, y_{<t}^c) \propto \sum_{i=1}^k 1_{y^c_t=v_i} \exp\left(\frac{-d(k_j, f_{\Theta^c}(x^c, y_{<t}^c))}{\tau}\right)$$

with $d(.,.)$ the euclidean distance between the child’s output representation and the key of the j^{th} nearest neighbor retrieved, τ is a temperature parameter, is an indicator function and $\propto$ is a ‘proportion to’ symbol.

Finally, the final probability distribution to predict next token y_t^c is obtained using a **linear interpolation** between the probability directly predicted by the child NMT model and the probability based on the k NN retrieval from the parent model’s datastore as:

$$p(y_t^c | x^c, y_{<t}^c) = \lambda p_{parent-kNN}(y_t^c | x^c, y_{<t}^c) + (1 + \lambda) p_{child-NMT}(y_t^c | x^c, y_{<t}^c)$$

with λ , a hyperparameter to be tuned accordingly.

3 Theoretical framework and Methodology

3.1 Theoretical Framework

The proposed research is based on the theoretical framework of **transfer learning in NMT**, with a meticulous interest in **inference-time knowledge transfer**. We opt for such a framework mainly because:

1. **Transfer learning** is the main base of our research, willing to improve the child’s performance by leveraging parent model’s knowledge during child’s initialization, training and inference.
2. **Non-Parametric Retrieval (k NN-MT, k NN-TL)** offers the possibility to query the parent datastore at child’s inference time.
3. **Confidence-Aware inference** refers to the child’s ability to dynamically decide whether additional external support is needed when uncertain.

No alternatives are explored as:

1. Static transfer learning techniques (e.g., fine-tuning and knowledge distillation) do not fully exploit the parent knowledge during child inference.
2. Data augmentation methods (e.g., back-translation) do enrich training data, which is essential in NMT, yet they fail at permitting real-time decision-making at decoding time.

At last, a framework, that combines dynamic transfer learning, non-parametric retrieval and confidence-aware gating, is deemed more flexible and efficient.

3.2 Methodology and techniques

1. Baseline Reproduction & analysis

- Reproduce and evaluate **k NN-TL framework** on a low-resource MT task (Moroccan dialect to MSA).
- Conduct an analysis to find out when and where the parent knowledge usage, at inference, helps or harms.

2. Design of the dynamic transfer learning framework

Introduce a **confidence-based routing approach** and explore different routing strategies, as follows:

- If the child’s confidence is below a set threshold, query the parent datastore.
- Else, rely on the child’s output.

3. Experiments

- **Datasets:** Work with low-resource MT datasets (e.g., PADIC[Meftouh et al., 2015], DODA², MADAR[Bouamor et al., 2019], etc).
 - **Models:** Implement effective and powerful pre-trained parent NMT models (e.g., trained on English-Arabic language pair) and low-resource child models (e.g., Moroccan dialect-Arabic).
 - **Metrics:** Assess translation quality (BLEU, BertScore) and efficiency in terms of inference time, memory usage and number of datastore queries.
 - **Baselines:** Compare obtained results with previous approaches (e.g., k NN-MT[Khandelwal et al., 2022], k NN-TL[Liu et al., 2023] and fine-tuned models).
4. **Generalization:** Conduct **ablation studies** to measure the possible impact of the confidence estimation and threshold tuning methods, as well as the size and quality of the parent datastore.

²<https://github.com/darija-open-dataset/dataset>

4 Preliminary Results

4.1 Small-scaled experiments:

- In order to measure the feasibility of the k NN-TL framework in a low-resource translation task, we have conducted series of exhaustive and controlled experiments translating from Moroccan dialect (Darija) to Modern Standard Arabic (MSA), using PADIC [Meftouh et al., 2015] as the child dataset (almost 5k for training, 1k for validation and 1k for testing) and OPUS-100 ³ as the parent dataset (almost 800k rows for training, 2k for validation and 2k for testing).
- *It is also important to note that given the restricted resources, mainly hardware and GPU power, we've unfortunately downsized the experiments scale (less training data, smaller models, smaller batch sizes, etc), all in order to see how the framework will perform under such distressing circumstances!*
- Regardless of the numerous configurations that were implemented and the extensive efforts that were made, **k NN-TL wasn't able to show any improvement for our task**, with BLEU scores at inference ranging from **0.02 to 0.83**.
- We've tried to give out hypothesizes for a such notable failure, as follows:
 1. **The linguistic complexity and the rich morphology of the Arabic language, as well as the insufficient training size (less than a million)** may be very plausible contributing factors to the 'not-so-great' parent model's performance, condemning the transfer learning via k NN-TL to fail, at advance.
 2. **The apparent domain-mismatch between parent and child data** is a well-known obstacle in transfer learning, which may not exclude k NN-TL as it relies strongly on parent representation-based alignment during child training. In fact, The parent data (Opus-100) is mostly formal, clean, and from structured domains (e.g., legal, news, subtitles) whereas PADIC is noisy and morphologically irregular. To that, The reverse parent model (ar $\rightarrow$ en), though strong in BLEU, produces low-quality pseudo-sources, highlighting the domain gap.
 3. **Poor quality of the pseudo-source data** which was conceived by back-translation the child target data using the reverse parent model (ar-en). Despite its reasonable BLEU score (37-39), the outputs were semantically noisy and low quality. It results in a possible weak alignment between pseudo and actual darija source data, a primary assumption in the k NN-TL's child training based on representation consistency learning.
 4. **Significant linguistc difference between the two source languages (Darija and English)**, that might failed the child's token-matching (TM-TL) [Aji et al., 2020] initialization; In addition to the child's initialization with copied parameters from the parent model like done in TL[Zoph et al., 2016], comes also the initialization of the its' encoder embedding layers by assigning the embeddings of common tokens from the parent model.

³<https://huggingface.co/datasets/Helsinki-NLP/opus-100>

4.2 Large-scaled experiments:

- Pursuing an unaltered implementation of k NN-TL shows to be a major necessity, in order to get a fair observation and a trusted measurement of the feasibility of the said framework in our low-resource MT task.
- Thus, Enlarging the scale of the experiments was heavily required such as:
 1. Increasing the parent training size. We opted for 5.6M randomly selected sentences of the HPLT dataset⁴. The validation was done on 3K sentences of the same dataset. The vocabularies are learned using the joint source-target BPE with 40K merge operations [Sennrich et al., 2016]
 2. Additional GPU power naturally follows, thus, we resorted to *kaggle*⁵ to exploit the 30 hours weekly offered resources per notebook, in particular the **two T4 GPUs** (0: capabilities = 7.5 ; total memory = 14.741 GB ; name = Tesla T4 — rank 1: capabilities = 7.5 ; total memory = 14.741 GB ; name = Tesla T4)
- As for the child data, we worked with with MADAR dataset [Bouamor et al., 2019] (8K for training, 2K for validation, 2K for testing), in addition PADIC [Meftouh et al., 2015].
- *Please note that with the low memory storage provided by kaggle (20Go), the inference part of the experiments had to dropped unfortunately; at the parent datastore creation, which is expected to have billions of entries given the large training size, only 5% was loaded before the forced abortion of the whole process due to memory saturation.*
- In spite of us going for an unchanged training configuration and working with a well-trained parent model (BLEU score: 56.16 at test time), **k NN-TL has failed to show-case any improvement for our task**, with child BLEU scores at validation time, standing at 6.64 for PADIC and 10.11 for MADAR, in comparison to children trained with TM-TL [Aji et al., 2020] (BLEU score peaked at 11.53 for MADAR, 9.01 for PADIC).
- Again, we're adding to hypothesizes mentioned above, so as to justify such a poor performance, the following:
 1. **The linguistic complexity and the rich morphology of the Arabic language as well as the Moroccan dialect** call for a careful and studied way to handle the child data and **investigate other tokenization and subword segmentation methods** besides Moses [Koehn et al., 2007] and BPE⁶ (Byte Pair encoding) as implemented in subword-NMT [Sennrich et al., 2016], as well as the use of **morphological information**.
 2. **VERY IMPORTANT:** In an effort to stay true to the original k NN-TL training configuration, we followed the same preprocessing approach described in their paper and github repository⁷ namely, binarizing the child data using the parent model's vocabulary. While this was effective for Turkish–English due to vocabulary and domain overlap, our case is a proof of a possible higher linguistic divergence between Darija (Moroccan dialect) and Modern Standard Arabic (MSA), and between (PADIC OR MADAR) and OPUS-100 domains.

⁴<https://opus.nlpl.eu/HPLT/ar&en/v1.1/HPLT>

⁵<https://www.kaggle.com/>

⁶<https://huggingface.co/learn/llm-course/chapter6/5>

⁷<https://github.com/NLP2CT/kNN-TL>

Consequently, a large fraction of **child tokens were out-of-vocabulary**, leading to **pervasive UNKNOWNTOKEN issues** in child training. This mismatch suggests that vocabulary reuse, while effective in some contexts, must be adapted with caution when dealing with dialectal or low-resource language pairs.

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